General Assembly

STUDY GUIDE

MEET THE DOM

The Basics

- DOM stands for **Document Object Model.** It's an interface that allows JavaScript to access and alter HTML elements on a web page.
    - Document: The web page, usually the HTML document
    - Object: Every HTML element is an object. These may also be referred to as nodes or elements.
    - Model: Often a hierarchical tree, called the DOM tree. This model describes how the objects are laid out on the page.

The DOM has four core capabilities:

- **Get Content** - The DOM can access content within a page, including the info a user has entered into a form field
- **Set Content** - The DOM can dynamically update the content of HTML elements on a page. For example, it can change the text of a heading or update an image when a user clicks a button.
- **Add Effects** - The DOM can add effects, like animations and fades, to HTML elements
- **Create Event Listeners** - Event listeners tell web page to "wait" or "listen" for a user's action before running a block of code.

DOM Manipulation - Getting and Setting Content

There are two main steps of DOM manipulation.

STEP ONE: "Grabbing"

- Finding and selecting the appropriate HTML element
- HTML and JavaScript don't talk directly to each other, so you have to use DOM methods to select an element to manipulate
- Two methods of grabbing:
    - document.querySelector( ): Finds the first matching element
    - document.querySelectorAll( ): Finds all matching elements
    - "document" tells JavaScript to look at the HTML document
    - the ( ) will contain information about the specific element you want to grab

STEP TWO: Manipulating the element

- Accessing, updating, or removing it
- Some simple manipulations:
    - getAttribute( ) will retrieve an attribute
    - setAttribute( ) will update an attribute
    - removeAttribute( ) will remove an attribute
    - document.createElement(tagName) can create a new element. It will work through the variable, not the HTML itself, but whatever you add will appear in the HTML.